geometries, e.g. handling multiple classes of ShapeNet [8] simultaneously. The attention mechanism at the core of our architecture has the potential to enforce local control of the interpolation, as seen in Figure 5. Further investigation is needed to explore the possibility to introduce additional priors on the attention to force a semantically meaningful localization and interpolation behavior. Our method shares a common drawback with most transformed-based architectures, requiring long training and post-processing time due to the nature of the refinement procedure. All code and data is publicly available 1. 7 Acknowledgment This work is supported by the ERC Grant No. 802554 (SPECGEO) and the MIUR under grant "Dipartimenti di eccellenza 2018-2022".

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